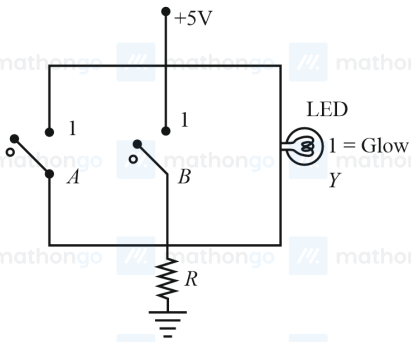


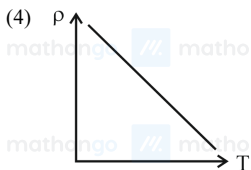
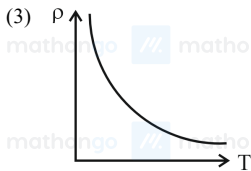
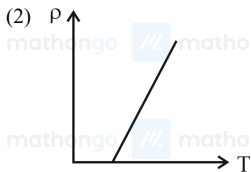
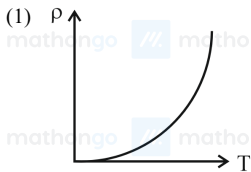
1. Name the logic gate equivalent to the diagram attached



[2023 (06 Apr Shift 1)]

- (1) NAND
(2) AND
(3) NOR
(4) OR
2. The resistivity (ρ) of semiconductor varies with temperature. Which of the following curve represents the correct behaviour?

[2023 (06 Apr Shift 1)]



3. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**

Assertion A: Diffusion current in a $p - n$ junction is greater than the drift current in magnitude if the junction is forward biased.

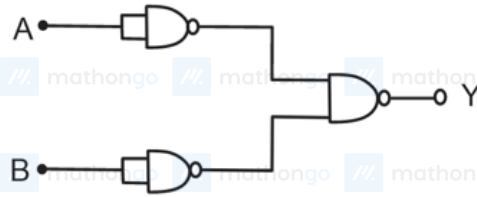
Reason R: Diffusion current in a $p - n$ junction is from the n -side to the p -side if the junction is forward biased.

In the light of the above statements, choose the most appropriate answer from the options given below.

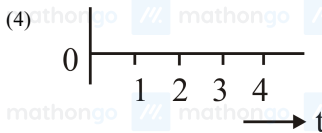
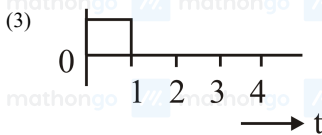
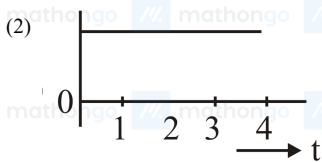
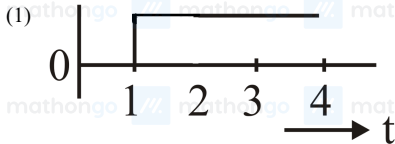
[2023 (06 Apr Shift 2)]

- (1) Both A and R are correct but R is NOT the correct explanation of A
(2) A is correct but R is not correct
(3) A is not correct but R is correct
(4) Both A and R is correct and R is the correct explanation of A

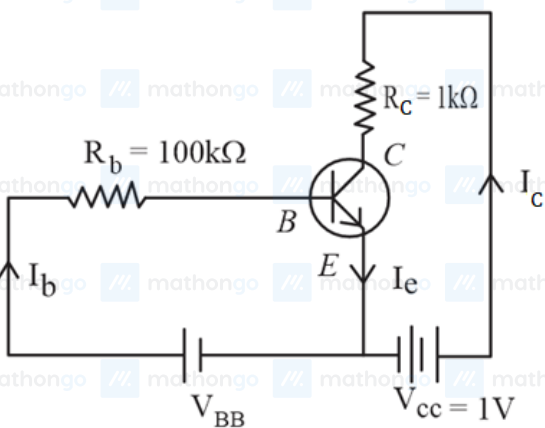
4. For the logic circuit shown, the output waveform at Y is



[2023 (08 Apr Shift 1)]



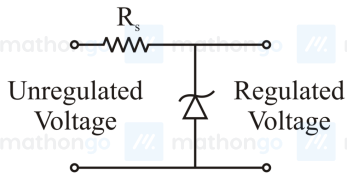
5. For a given transistor amplifier circuit in CE configuration $V_{CC} = 1\text{ V}$, $R_C = 1\text{ k}\Omega$, $R_b = 100\text{ k}\Omega$ and $\beta = 100$. Value of base current I_b is



[2023 (08 Apr Shift 2)]

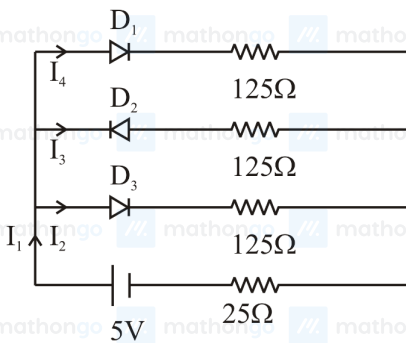
- (1) $I_b = 1.0\text{ }\mu\text{A}$
 (2) $I_b = 0.1\text{ }\mu\text{A}$
 (3) $I_b = 100\text{ }\mu\text{A}$
 (4) $I_b = 10\text{ }\mu\text{A}$

6. A zener diode of power rating 1.6 W is to be used as voltage regulator. If the zener diode has a breakdown of 8 V and it has to regulate voltage fluctuating between 3 V and 10 V. The value of resistance R_s for safe operation of diode will be



[2023 (10 Apr Shift 1)]

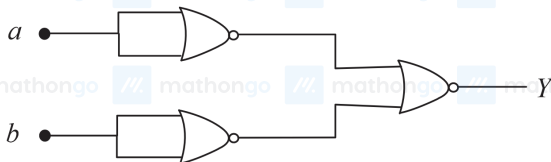
- (1) 10 Ω
(2) 12 Ω
(3) 13.3 Ω
(4) 13 Ω
7. If each diode has a forward bias resistance of 25 Ω in the below circuit,



Which of the following options is correct?

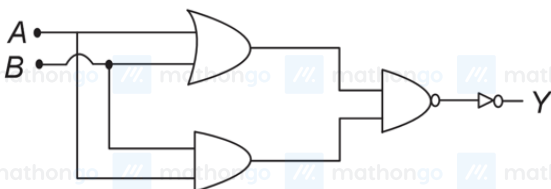
[2023 (10 Apr Shift 2)]

- (1) $\frac{I_1}{I_2} = 1$
(2) $\frac{I_2}{I_3} = 1$
(3) $\frac{I_1}{I_2} = 2$
(4) $\frac{I_3}{I_4} = 1$
8. The logic performed by the circuit shown in figure is equivalent to



[2023 (11 Apr Shift 1)]

- (1) AND
(2) NOR
(3) OR
(4) NAND
9. The logic operations performed by the given digital circuit is equivalent to:



[2023 (11 Apr Shift 2)]

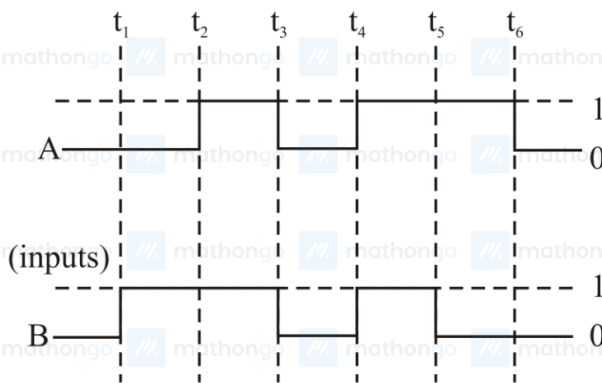
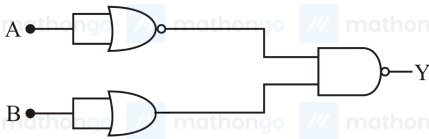
- (1) NOR
(2) AND
(3) OR
(4) NAND

10. In an n-p-n common emitter (CE) transistor the collector current changes from 5 mA to 16 mA for the change in base current from $100\text{ }\mu\text{A}$ and $200\text{ }\mu\text{A}$, respectively. The current gain of transistor is ____.

[2023 (12 Apr Shift 1)]

- (1) 110
(2) 210
(3) 0.9
(4) 9

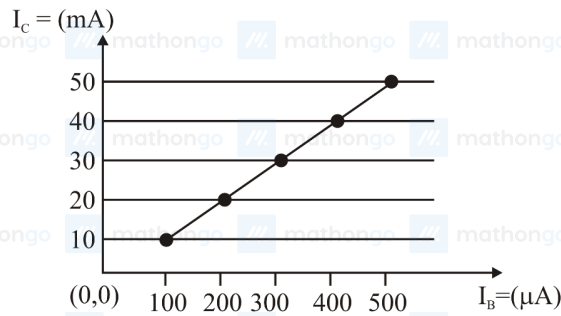
11. For the following circuit and given inputs A and B , choose the correct option for output ' Y '



[2023 (13 Apr Shift 1)]

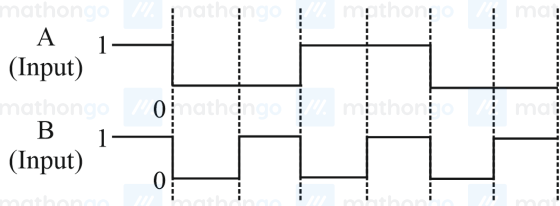


12. From the given transfer characteristic of a transistor in CE configuration, the value of power gain of this configuration is 10^x , for $R_B = 10 \text{ k}\Omega$, and $R_C = 1 \text{ k}\Omega$.
The value of x is _____.



[2023 (13 Apr Shift 1)]

13. The output from a NAND gate having inputs A and B given below will be,



[2023 (13 Apr Shift 2)]

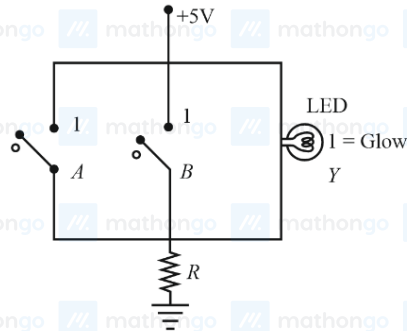
- (1)
- (2)
- (3)
- (4)

ANSWER KEYS

1. (3) 2. (3) 3. (2) 4. (1) 5. (4) 6. (1) 7. (3) 8. (1)
9. (2) 10. (1) 11. (1) 12. (3) 13. (2)

1. (3)

Let's assume that when the path is short, the state is called state 1 and when it is open, the state is called state 0.



Depending up on whether the path is short or not in the circuit, the following table can be obtained:

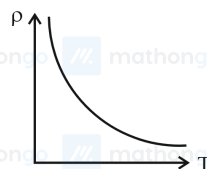
A	B	Out
1	1	0
1	0	0
0	1	0
0	0	1

The above table is similar to the truth table corresponding to NOR gate. Hence, the circuit indicates a NOR logic gate.

2. (3)

When a modest amount of energy is applied to semiconductors, electrons quickly travel to the conduction band and are then free to move across the crystal's lattice.

If thermal energy is provided, the conduction of electrons will increase. Hence, as the temperature is increased, the resistivity of the semiconductor is decreased according to the following diagram:



3. (2)

The electric current due to the charge concentration gradient is known as Diffusion current. When the charge carriers move in a semiconductor due to the applied electric field, the resulting current is known as drift current. If the positive terminal of the battery is connected to the p-type while the negative terminal of the battery is connected to the n-type, the movement of electrons is eased as they can travel to the positive terminal of the battery. This causes an increase in the diffusion current. Hence, the diffusion current is more than the drift current. So, A is correct

In an n-type semiconductor, the concentration of electrons is more than that of the holes. While in a p-type semiconductor, the concentration of holes is more than that of the electrons. During the formation of a p-n junction diffusion and drift are the two processes taking place. When a p-n junction is formed, holes diffuse from the p-side to the n-side while electrons diffuse from the n-side to the p-side. This result due to the concentration gradient across p and n sides, which gives rise to a diffusion current across the junction. The diffusion results in an electric field from p-side to n-side. Thus, R is incorrect as diffusion current in p - n junction is from p side to n-side.

4. (1)

All the gates used in the diagram represent NAND gates.

Thus, the output Y can be calculated as follows

$$\begin{aligned} Y &= \overline{\overline{A} \cdot \overline{B}} \\ &= \overline{\overline{A} + \overline{B}} \\ &= A + B \end{aligned}$$

The above equation suggests that the output of the given combination yields the same result as an OR gate.

The truth table can be written as follows

A	B	$\overline{A}$	$\overline{B}$	$Y = \overline{\overline{A} \cdot \overline{B}}$
0	0	1	1	0
0	1	1	0	1
1	0	0	1	1
1	1	0	0	1

Hence, this is the correct output.

5. (4)

Considering the transistor in saturation mode, $V_{CE} = 0$

Using Kirchhoff's voltage rule,

$$-I_C R_C + V_{CC} = 0$$

$$\Rightarrow I_C = \frac{V_{CC}}{R_C}$$

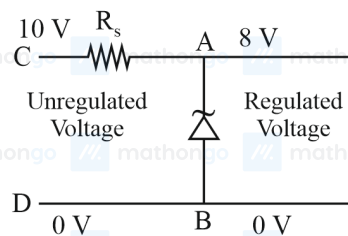
$$\Rightarrow I_C = \frac{1\text{ V}}{10^3 \Omega} = 10^{-3} \text{ A}$$

So,

$$\beta = \frac{I_C}{I_B} = \frac{10^{-3} \text{ A}}{I_B}$$

$$\Rightarrow I_B = \frac{10^{-3}}{10^2} = 10^{-5} \text{ A} = 10 \mu\text{A}$$

6. (1)



Power rating of zener diode = 1.6 W

From the diagram above,

$$V_A - V_B = 8 \text{ V}$$

Power is given by

$$P = Vi$$

$$\Rightarrow i = \frac{1.6 \text{ W}}{8 \text{ V}} = 0.2 \text{ A}$$

$$\text{Also, } V_C - V_A = 10 \text{ V} - 8 \text{ V} = 2 \text{ V}$$

Thus, using Ohm's law,

$$R_s = \frac{2 \text{ V}}{0.2 \text{ A}} = 10 \Omega$$

7. (3)

As the diode D_2 is connected in reverse order, there is no current through it.

The effective resistances of the circuit can, then, be calculated as follows:

$$R_{eff} = 25 + \frac{(25+125)^2}{(25+125) + (25+125)} \Omega$$

$$= 100 \Omega$$

Each current through the diodes can be calculated as follows:

$$I_1 = \left(\frac{5 \text{ V}}{100 \Omega} \right)$$

$$= 0.05 \text{ A}$$

As the diode D_2 is non-conducting, the current I_1 will be divided equally among the other two diodes D_1 and D_3 .

Hence, the currents in the diodes can be written as

$$I_2 = \frac{0.05 \text{ A}}{2}$$

$$= 0.025 \text{ A}$$

$$I_4 = 0.025 \text{ A}$$

Thus,

$$\frac{I_1}{I_2} = \frac{0.05 \text{ A}}{0.025 \text{ A}}$$

$$= 2$$

8. (1)

All the gates used in the given configuration are NOR gates. Thus, the output can be found as follows-

$$Y = \overline{(\overline{a} + \overline{b})}$$

$$= \overline{(\overline{ab})}$$

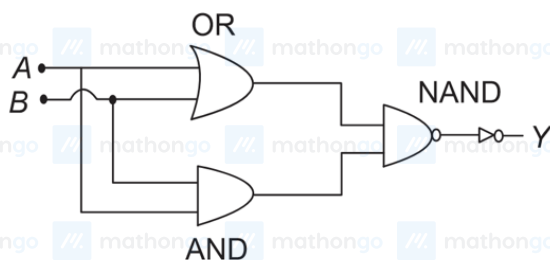
$$= ab$$

The output of the given configuration can also be verified by the following truth table:

a	b	Output from first gate	Output from second gate	Y
0	0	1	1	0
0	1	1	0	0
1	0	0	1	0
1	1	0	0	1

The calculation and the truth table clearly shows that the configuration represents AND gate.

9. (2)



As shown in the figure the various logic gates have been mentioned.

Let the output before the NOT gate be Z .

Using the expression for the logic gates, it can be written

$$Z = \overline{(A + B) \cdot (A \cdot B)}$$

Hence, the output Y is

$$Y = \overline{Z} = \overline{\overline{(A + B) \cdot (A \cdot B)}}$$

$$\Rightarrow Y = A \cdot B$$

Hence the digital circuit is equivalent to an AND gate.

10. (1)

The change in collector current for the given transistor can be written as

$$\Delta I_c = 16 \text{ mA} - 5 \text{ mA}$$

$$\Delta I_c = 11 \text{ mA}$$

The change in base current for the given transistor can be written as

$$\Delta I_b = 200 \mu\text{A} - 100 \mu\text{A}$$

$$= 100 \mu\text{A}$$

The formula to calculate the current gain in CE configuration is given by

$$\beta = \frac{\Delta I_c}{\Delta I_b} \dots (1)$$

Substitute the values of the known parameters into equation (1) to calculate the required quantity.

$$\beta = \frac{\Delta I_c}{\Delta I_b}$$

$$= \frac{11 \text{ mA}}{100 \mu\text{A}}$$

$$= 110$$

11. (1)
The signal A is given to a NOR gate and B is given to an OR gate. The output of these two signals is given to a NAND gate. Using the laws of Boolean algebra,

$$Y = \overline{\overline{A} \cdot B} = A + \overline{B}$$

Analysing from the input graphs, one can see,

$$t_1 - t_2 : A = 0, \overline{B} = 0, Y = 0$$

$$t_2 - t_3 : A = 1, \overline{B} = 0, Y = 1$$

$$t_3 - t_4 : A = 0, \overline{B} = 1, Y = 1$$

$$t_4 - t_5 : A = 1, \overline{B} = 0, Y = 1$$

$$t_5 - t_6 : A = 1, \overline{B} = 1, Y = 1$$

12. (3)
The formula to calculate the current gain (β) in CE configuration is given by

$$\beta = \frac{\Delta I_c}{\Delta I_b} \dots (1)$$

Considering the third and the first points on the given graph, the current gain can be calculated as follows:

$$\beta = \frac{(30 - 10) \times 10^{-3} \text{ A}}{(300 - 100) \times 10^{-6} \text{ A}}$$

$$= 100$$

The formula to calculate the power gain (P) in CE mode can be written as

$$P = \beta^2 \cdot \frac{R_c}{R_b} \dots (2)$$

Substitute the known values of the parameters into equation (2) and solve to calculate the required unknown parameter.

$$10^x = 100^2 \times \frac{1 \text{ k}\Omega}{10 \text{ k}\Omega}$$

$$= 1000$$

$$= 10^3$$

$$\Rightarrow x = 3$$

13. (2)

The output of NAND gate is

$$Y = (AB)'$$

The truth table is

$$\begin{array}{ccc} A & B & Y \end{array}$$

$$\begin{array}{ccc} 0 & 0 & 1 \end{array}$$

$$\begin{array}{ccc} 0 & 1 & 1 \end{array}$$

$$\begin{array}{ccc} 1 & 0 & 1 \end{array}$$

$$\begin{array}{ccc} 1 & 1 & 0 \end{array}$$

Y would be zero only when both A and B are 1. Clearly, for all other input conditions the output will be 1. Hence, this is the right option.